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High-Dimensional Signal Processing using Classical-Quantum Machine Learning Pipelines with the TensorFlow Stack, Cirq-NISQ, and Vertica

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***Abstract*—Modeling High Dimensional Signal Data (HDSD) is challenging due to statistical complexities and an imbalance of noise to signal data, which creates ground truth problems. Likewise, enabling recognition and detection analytics on the information-bearing portion of the noise-obscured HDSD is computationally cost-prohibitive due to the unprecedented size, speed, and scale of HDSD. Consequently, human experts often out-perform Deep Learning (DL) algorithms at HDSD recognition. This is problematic due to the spectrum of applications reliant on accurate HDSD detection. These include multi-sensor and biologic signal detection, virus outbreak surveillance, genetic association identification, image, audio, and sonar processing, source separation for speech identification, astrophysical source detection, financial time series, and satellite payload processing. To solve this problem and facilitate favorable DL outcomes, synthetic data possessing enough positive samples is needed to balance the HDSD dataset. Research reveals an entangled Quantum Generative Adversarial Network (QGAN) utilizing parameterized quantum circuits for the QGAN Generator (QGEN) and Discriminator (QDIS) exhibit an exponential advantage over a classical-GAN when generating highly accurate and cost-effective synthetic data. Using this method, an unsupervised TensorFlow-Quantum QGAN is prepared from Cirq Noisy Intermediate-Scale Quantum (NISQ) circuits. The QGEN and QDIS create new statistically consistent synthetic HDSD from real data; thereby solving the ground truth problem. Next, TensorFlow-GPU facilitates classical-DL to train the new synthetic HDSD dataset. Once validated, the TensorFlow model is exported to the production Vertica analytics tier for fast and accurate ML inference, where the HDSD is instantly identified and classified at ingest.**

Keywords— TensorFlow Quantum, Cirq, Entangled Quantum Generative Adversarial Network, Vertica Machine Learning, High-Dimensional Signal Data Processing

I. Introduction

Modeling High-Dimensional (HD) signal data is challenging due to statistical complexities and an imbalance of noise to signal data, which creates ground truth problems. For this reason, enabling recognition and detection analytics on the information-bearing portion of these noise-obscured HD signals is computationally cost-prohibitive due to their unprecedented size, speed, and scale. As a result, human experts typically out-perform deep learning algorithms at HD signal recognitions. A wide spectrum of applications rely on HD signal detection [4], such as, multi-sensor radar detection, disease outbreak surveillance, genetic association identification, image, audio, and sonar signal processing, source separation for speech identification, astrophysical source detection, financial time series, bio-signal data augmentation, and satellite payload processing, to name a few. This problem can be solved by generating synthetic HD signals possessing a sufficient number of positive signal samples, thereby balancing the signal dataset and facilitating favorable Deep Learning (DL) outcomes. An unsupervised Quantum Generative Adversarial Network (uQGAN) leveraging parameterized quantum circuits for both the generator and discriminator has been shown to exhibit an exponential advantage over its classical GAN counterpart in generating synthetic data, indistinguishable from the real data with a high degree of accuracy. Prior research [3] leveraged an entangled uQGAN prepared from Cirq Noisy Intermediate-Scale Quantum (NISQ) circuits using TensorFlow Quantum (TFQ). Likewise, [2] reveal success in creating quantum data from real data using TFQ quantum generator (QGEN) and quantum discriminator (QDIS) objects. This research aims to solve ground truth problems that plague high-dimensional signal data by using an entangled and unsupervised Quantum Generative Adversarial Network (uQGAN) to create statistically accurate synthetic data for classical modeling.

II. Methodology

Combing the work of [1,2,3] and using them as a template, real signal data is first balanced and converted to binary on classical computing hardware prior to quantum state encoding on a Quantum Processing Unit (QPU). Efficient dataset superposition generation occurs using the entangled TFQ uQGAN by approximating the Quantum Random Access Memory (QRAM) from a shallow Cirq-NISQ circuit. The QGEN then creates synthetic signal data from the real data. To obfuscate decoherence effects, a noise model with the three most common quantum errors (bit-flip, phase-flip, and bit-and-phase-flip) will be integrated and used by the QGEN to train the synthetic data model, making the QGEN's data model robust to NISQ decoherence by improving active Quantum Error Correction (QEC). Both real and synthetic data are entangled at the QDIS, creating new "ground truth" data which is then used to train a classical TensorFlow (TF) Convolutional Neural Network (CNN).

The validated TF/model is then exported to a Vertica production cluster where two inference options are presented. First, high-speed Machine Learning (ML) inference can be executed against the imported TF/model using live sensor data ingested into Vertica. The second option leverages model-stacking techniques where the TF feature map is extracted from the DL model and used to create a VerticaML classification, regression, or cluster model. The benefits of DL/ML model stacking include enhanced model performance and model explainability for AI governance. High-speed inference is then performed against the newly created VerticaML model using live signal data at ingest.

Sample dataset structure:

- The Galaxy Zoo dataset has 61,578 training datapoints (images) and 79,975 test datapoints.

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- All images are JPGs and are 424x424 pixels in size.
- The first column of the training_solutions_rev1.csv comprises the training labels (GalaxyID), which corresponds to each training image datapoint.
- The next 38 columns (features) in training_solutions_rev1.csv file consist of a flattened array of pixel vectors collapsed into a single row.
- Upon converting the datasets from RGB to Greyscale, the pixel values will range from 0 to 255 where 0 corresponds to white and 255 corresponds to black and any value between 0 and 255 corresponds to a color which is a mix of black and white.
- The dimension of galaxy zoo dataset is 38, too large to assign individual qubits. As such, the dataset will be reduced to 1,000 images total and resized to 4x4 pixels to accommodate QPU processing.

A. Mapping Classical Data to a Quatum State

Quantum RAM hardware and circuitry used to read classical data into a NISQ system for quantum processing. In order to efficiently load classical data into the NISQ system, the data must be sized appropriately for the QRAM. As such, the QRAM often becomes the limiting factor in classical-quantum data conversions. This work leverages basis encoding, which turns vectorized data into binary strings prior to creating superpositions of those basis states in the quantum computer's QRAM. Once the classical data is successfully loaded into the NISQ system, the qubit states represents the classical data. Entangled QGANs have experienced sucess in approximating the QRAM upon learning a shallow quantum circuit. This efficiently generates a superposition of a small balanced, classical dataset like the one used in this work..

B. Quantum Generative Adversarial Network

Entanglement extends GAN's applicability to quantum, overcoming the classically hard probability distributions that preclude many classical-GAN models from converging or reaching a state where the model prediction loss approaches a Nash equilibrium. Unlike a classical-GAN where the generator supplies the discriminator with either real or fake data, the quantum generator in an entangled-GAN provides the quantum discriminator with both real and fake data entangled together. This novel technique facilitates model convergence to a stable state on problems that classical and hybrid quantum-classical GAN models previously failed on. The entangled QGAN used in this work is unsupervised with the QDIS learning from unlabeled QGEN data.

C. Classical Deep Learning

Due to the many millions of classical datapoint features contained in the Galaxy Zoo dataset, a classical Neural Network will outperform any quantum neural network. For this reason, the synthetic ground truth dataset will be moved from TFQ to TF-Keras on classical hardware for training. The transition is seamless since TFQ integrates Cirq-quantum computing algorithms and logic with primitives compatible with contemporary TensorFlow APIs. The classical CNN used in this work is premised on a pre-trained ResNet18 deep residual neural network. The ResNet architecture leverages seven stages of downsampling, with each stage doubled until a maximum depth of 512 is reached. Depending on whether model stacking techniques will be used (with VerticaML), the final softmax output (classifier) CNN layer may or may not be removed.

D. Vertica Machine Learning

Vertica is a Massively Parallel Processing (MPP) Analytics platform built from the ground up to handle massive petabyte-scale workloads. Touting a balanced, distributed, compressed, columnar paradigm, Vertica further integrates many of the most commonly used ML algorithms for data preparation, exploration, and model evaluation, which are fully optimized for distributed cluster systems. By managing ML models as firstclass citizens, Vertica eliminates many barriers to ML at high-scale. In addition to AI governance protocols, Vertica's in-database ML features provides many built-in ML functions, such as classification, clustering, and prediction using k-means, bisecting k-means, linear regression, logistic regression, support vector machines, random forest, naïve Bayes, and many other integrated Vertica ML algorithms.

E. Model Stacking

Model stacking uses more than one model for predictions. In this work, the output from the TF-CNN model is used as the input for a higher-level VerticaML meta model. The selected VerticaML Model (regression, classification, clustering) learns the input predictors and offers a better response output prediction. Model stacking often achieves higher prediction accuracy since the model learners all contribute to different model strengths. Vertica is an production analytics platform where TF/models can be directly imported for inference, or TF/model output can be used as input for Vertica's in-database ML algorithms. Either method avails model scoring against live ingest, in real-time.

III. RESULTS

This section presents the entire hybrid classical-quantum machine learning pipeline for high-dimensional signal processing analytics. First, the high-dimensional signal data is described, along with its quantum state preparation. Next, the entangled uQGAN model used for synthetic signal data creation is explored. Lastly, the classical training method is detailed, though these final steps are still in progress as of the time of this writing.

A. Classical Data and Quantum State Preperation

The Open Source Kaggle Galaxy Zoo image training dataset was balanced and annotated, with all signal-noise removed. Galaxy images were mutated from color (RGB) to greyscale and resized to 4x4 on a classical system. This size accommodates QPU processing on existing NISQ systems.

- Images were next converted into a Two-Dimensional (2D) array of pixel vectors.
- The 2D matrix was then encoded in binary strings in preparation for QRAM loading of classical data in superposition.
- Encode the classical data as quantum circuits by representing each pixel with a qubit where the quantum state is dependent on the pixel's value.
- Qubits with pixel indices values exceeding a preset threshold were rotated through an X gate, thereby creating Cirq-NISQ circuits.

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- TFQ tensors are created from the Cirq-circuits. This model circuit used the same X gate for all (n) TFQ layer instances.
- A Keras model was wrapped around the Cirq-TFQ model objects in preparation for feeding the newly created (encoded classical) quantum data to the uQGAN's QGEN.

B. *Synthetic Data Generation using an Entangled uQGAN*

An adversarially trained entangled uQGAN efficiently and cost-effectively trained the QGEN and QDIS algorithms to synthesize new signal data by estimating the real signal data's density and sampling that estimation.

- Two Parametrized Quantum Circuits on Google's NISQ system were used to train the TFQ uQGAN model. The QGEN circuit was used to create the new synthetic quantum data.The QDIS circuit was them used for the data predictions.
- A TFQ-keras layer was used to share variables between the two QGEN and QDIS quantum circuits.
- The QGEN continually fed the QDIS a signal data sample, and the QDIS determined if the sample was real or synthetic data. The QGEN then updated its model after each correct QDIS selection, making it more difficult for the QDIS to guess accurately.
- The uQGAN's continual retraining the QGEN model to fool the QDIS afforded a new, high-quality ground truth signal dataset for classical training.

NISQ systems are error-prone inhibiting practical quantum uses, like this work proposes. As such, future uQGAN work will include a noise model with common bit-flip, phase-flip, and bit-and-phase-flip errors built into the QGEN's synthetic dataset. The QDIS wil then determine if the sample has one of the three quantum errors. The integration of this custom quantum noise model will improve QEC robustness and facilitate a quantum advantage for practical use cases

C. *Model Training and Inference (In progress)*

The new ground truth dataset is anticipated to improve signal classification performance; however, this work is still under development as of the time of this writing. Upon training a TF-keras CNN model from the new ground truth dataset, the model will be exported to tbe Vertica production tier. Once loaded into Vertica, inference can be performed against the TF/model directly, or a DL/ML model stacking technique can be leveraged to improve model performance and accommodate AI governance explainability protocols. Using model stacking, the TF/model's feature map is extracted into a VerticalML clustering, regression, or classification model. In this situation, inference is also performed in real-time against sensor data ingest; however, model scoring occurs against the new VerticaML (cluster, regressor, or classifier) model instead of the imported TF/model.

IV. CONCLUSION

Motivated by DL algorithm detection and recognition limitations in high-dimensional signal analytics, this work explored a computationally cost-effective option for generating synthetic signal data from the information-bearing portion of noise-obscured signal datasets. This research revealed a new method for overcoming the ground truth problems that cripple high-dimensional signal data. Classical data was first converted to binary prior to quantum state preparation. Next, an entangled uQGAN created statistically accurate synthetic data using a custom quantum error noise model to improve active NISQ error correction. Synthetic and real signal data was entangled at the uQGAN discriminator where a resultant ground truth dataset was created. Final steps include training a classical TF/CNN on GPUs, with the validated model exported to Vertica for ML model stacking and/or TF inference against live production signal data.

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